

CS 101 Advanced Lecture Notes: Deep Neural Architecture

Section 1: Transformer Attention Mechanism

The core mechanism behind modern large language models is multi-head self-attention.

Attention computes queries (Q), keys (K), and values (V) using linear projections.

The scaled dot-product attention formula is $\text{Attention}(Q, K, V) = \text{softmax}(Q * K^T / \sqrt{d_k}) * V$.

This allows models to capture long-range token dependencies in parallel across sequences.

Section 2: Quantization and Edge Inference

Model quantization reduces numerical precision of weights from FP32 or FP16 down to INT8 or INT4.

Techniques such as AWQ (Activation-aware Weight Quantization) and GPTQ minimize perplexity degradation while shrinking memory footprint by 75%, allowing local execution on commodity hardware.

The course practical lab requires implementing INT8 dynamic quantization in PyTorch.